CS 255 Business Requirements Document

System Components and Design

Purpose

- The client is DriverPass., Liam is the Owner, Ian is his IT Officer.
- They want their system to assist potential customers pass their drivers test by providing online training, and on-the-road training if requested.

System Background

- System should be able to access data from a computer or mobile device.
- Needs to be able to download reports and other data.
- Ian needs full Admin control to aid in customer service, and manage internal access controls.
- Different employees need different access roles.
- Needs to be able to generate activity reports for reservations that link to the user: Who made it? Who canceled it? Who modified it last?
- Capability of tracking customer pairing with drivers/vehicles.
- System is web-based and on a cloud service.
- IT Office where the admin can control access.
- Secretary account where the secretary can schedule appointments by phone or walk-in.
- User account where students can schedule themselves.
- Driver that can log notes and teach the students.

Objectives and Goals

WHAT THE SYSTEM SHOULD BE ABLE TO DO:

Objectives:

- Track drivers and cars used for driving lessons.
- Track users' statuses.
- Automate scheduling.

Goals:

- Scheduling services for driving lessons.
- Online classes offering both materials and quizzes.
- Tracking ability where all operations made to the system are logged.
- Flexibility offering users the ability to register for their courses either online or through an agent at the company.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system will run on a cloud service web-based environment, should only take a few seconds to run and should update instantly/as changes are made.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The desire is for the system to be accessed via mobile or PC, so it should run on (but not limited to) Windows, Mac, Linux, Android, iOS devices. Should be browser agnostic.
- A database will be required to retain account, user, and other essential business information.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- There should be a separate login portal for employees and users.
- Username will not be case sensitive, but will measure against the database to prevent duplicates.
- System should inform the admin of problems as they occur with essential information.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Only the owner can make changes to the users on the employee side such as add/remove/modify.
- The system should query the database for changes at consistent, low volume use times (such as 02:00) and implement changes then for non-urgent updates.
- There should also be circumstances that force the system to update such as termination, username update, personal information update, etc.
- There should be a method to manually push through updates to the system.
- IT Admin needs access to add/modify users including customers and employees.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- User is required to have a password, and a form of multi-factor authentication setup. (Email, phone, app)
- The system should encrypt data while communicating. (HTTPS has this naturally built in.)
- Accounts should be locked after 3-5 failed login attempts, with a method to unlock by contacting and validating the account owner.
- System should require users to "register a device" that they use to login every 30-days similar to Brightspace, that way if a new device is detected, the user and admins can be notified of a potential breach.
- Users should be able to reset password using MFA.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

The system shall:

- Verify user information before logging in.
- Differentiate the type of user.
- Allow access from any browser.
- Track customer pairings with drivers/vehicles.
- Generate activity reports.
- Update with DMV information as they make updates.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- Different users are Customers and Employees with different tiers.
- Customer needs are: Able to check their online test progress, manage their special needs, see their student and teacher photos, select packages, update personal information, review driver notes, manage payment information and make payments.
- Employees can: Update user information, scheduler users for their tests and packages, review driver notes and other customer account informatin.

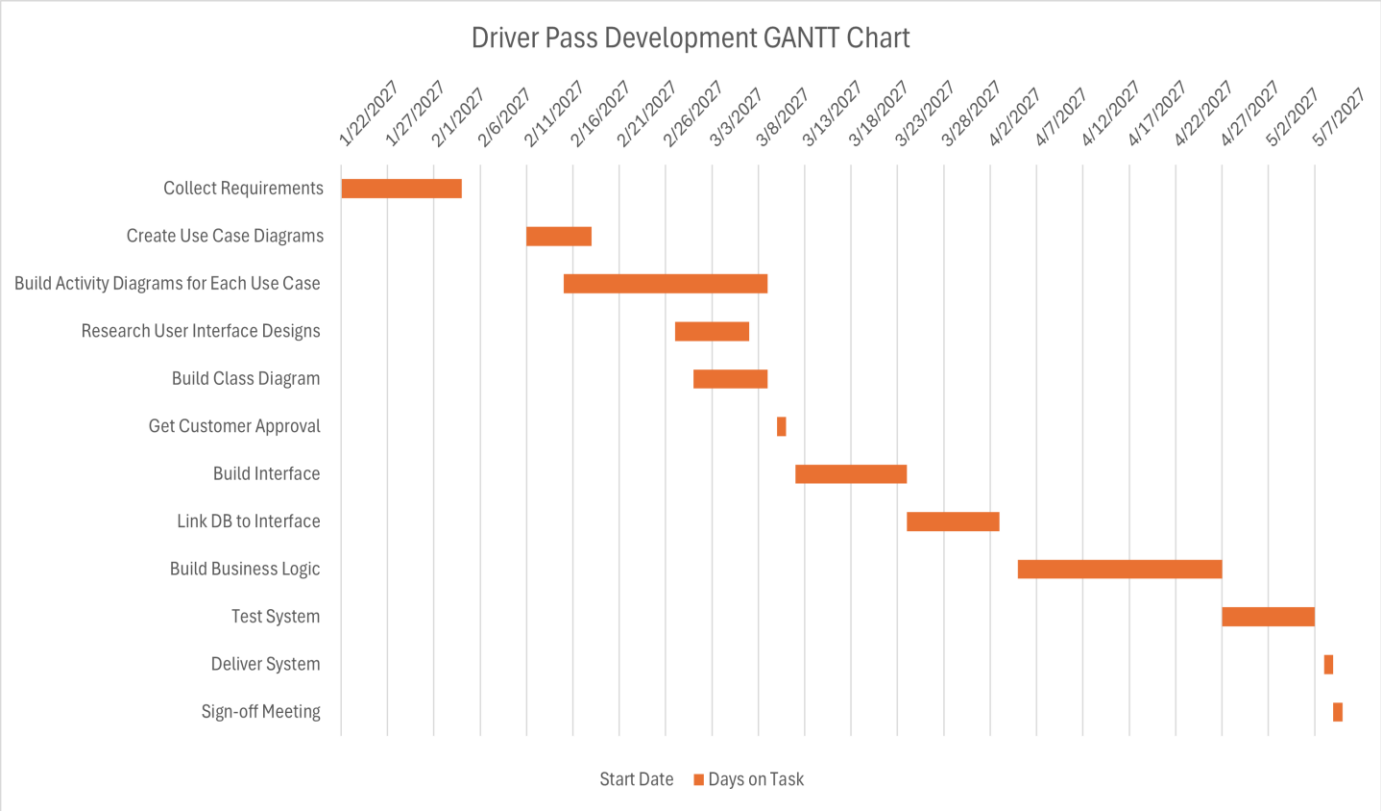
Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Users have access to the web.
- Users have basic computer literacy skills.
- Users have a phone or PC that they have everyday access to.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Meeting DriverPass requirements within the budget and time frame.
- Limited number of cars and packages.

Gantt Chart